Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	15 October 2026
Team ID	23891A05C2
Project Name	Medical Diagnosis Center
Maximum Marks	4 Marks

Technical Architecture:

The proposed Medical Diagnosis Center application is developed using the ServiceNow platform. Patients can use the Service Portal to register, browse diagnostic tests, book appointments, track test status, and access completed reports. Authorized staff manage appointments, update test progress, and maintain diagnostic records.

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	Provides the portal for patient registration, test catalog, appointment booking, and report viewing	ServiceNow Service Portal, HTML, CSS, JavaScript
2.	Application Logic-1	Handles patient registration, validation, and patient record management.	ServiceNow platform, Client Scripts, Business Rules, GlideRecord
3.	Application Logic-2	Manages appointment booking, slot validation, cancellation, and status updates.	ServiceNow Business Rules, Script Includes, GlideRecord
4.	Application Logic-3	Automates appointment updates, notifications, and test-status workflows.	ServiceNow Flow Designer, Notifications
5.	Database	Stores patient details, appointments, diagnostic tests, and reports.	ServiceNow application tables
6.	Cloud Database	Persistent storage managed by the ServiceNow platform.	ServiceNow-managed platform database
7.	File Storage	Stores uploaded diagnostic report files and related attachments.	ServiceNow Attachments

8.	External API-1	No external API is required for the core project functionality.	Not applicable
9.	External API-2	No external API is required for the core project functionality.	Not applicable
10.	Machine Learning Model	The application focuses on diagnostic workflow management; no ML model is required.	Not applicable
11.	Infrastructure (Server / Cloud)	Hosts the application, portal, database, and workflows.	ServiceNow cloud platform; browser-based access

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	Uses standard web technologies within the ServiceNow Service Portal. No separate open-source application framework is required for the core implementation.	HTML, CSS, JavaScript
2.	Security Implementations	Uses role-based access, authentication, table permissions, and Access Control Rules to protect patient and diagnostic records.	ServiceNow ACLs, Roles, Authentication
3.	Scalable Architecture	The application uses modular tables, reusable scripts, and workflows to support additional patients, tests, appointments, and reports.	ServiceNow Platform, Application Tables, Flow Designer
4.	Availability	The application runs on the ServiceNow-hosted platform and is accessible when the instance is available.	ServiceNow cloud infrastructure
5.	Performance	Uses structured database tables, appropriate queries, reusable logic, and efficient portal components to support routine operations.	GlideRecord, Service Portal, ServiceNow platform

References:

- C4 Model — Architecture diagram concepts
- IBM Cloud Architecture — IBM architecture reference
- AWS Architecture — AWS architecture reference
- ServiceNow Documentation — Official ServiceNow platform documentation